

# INTEGRAL EQUATIONS

## 21.1 INTRODUCTION

With the exception of the integral transforms of Chapter 20, we have for the most part been considering relations between an unknown function  $\varphi(x)$  and one or more of its derivatives. We now proceed to investigate equations containing the unknown function within an integral. As with differential equations, we shall confine our attention to linear relations, which are called **linear integral equations**. These integral equations are classified in two ways:

- If the **limits of integration are fixed**, we call the equation a **Fredholm** equation; if **one limit is variable**, it is a **Volterra** equation.
- If the **unknown function** appears **only under the integral** sign, we label it **first kind**. If it appears both **inside and outside** the integral, it is labeled **second kind**.

Here are some examples of these definitions. In each of the following equations,  $\varphi(t)$  is an unknown function whose value we seek.  $K(x, t)$ , which we call the **kernel**, and  $f(x)$  are assumed to be known. When  $f(x) = 0$ , the equation is said to be **homogeneous**.

This is a **Fredholm equation of the first kind**,

$$f(x) = \int_a^b K(x, t)\varphi(t) dt. \quad (21.1)$$

Next we have a **Fredholm equation of the second kind**, which is an eigenvalue equation with  $\lambda$  the eigenvalue,

$$\varphi(x) = f(x) + \lambda \int_a^b K(x, t)\varphi(t) dt. \quad (21.2)$$

Here we have a **Volterra equation of the first kind**,

$$f(x) = \int_a^x K(x, t)\varphi(t) dt; \quad (21.3)$$

and a **Volterra equation of the second kind**,

$$\varphi(x) = f(x) + \int_a^x K(x, t)\varphi(t) dt. \quad (21.4)$$

Why do we bother about integral equations? After all, the differential equations have done a rather good job of describing our physical world so far. However, there are several reasons for introducing integral equations.

First, we have placed considerable emphasis on the solution of differential equations **subject to particular boundary conditions**. For instance, the boundary condition at  $r = 0$  determines whether the Neumann function  $Y_n(r)$  is present when Bessel's equation is solved. The boundary condition for  $r \rightarrow \infty$  determines whether  $I_n(r)$  is present in our solution of the modified Bessel equation. To the contrary, an integral equation relates the unknown function not only to its values at neighboring points (derivatives) but also to its values throughout a region, including the boundary. In a very real sense the boundary conditions are built into the integral equation rather than imposed at the final stage of the solution. It will be seen later in this section that if we construct an integral equation that is equivalent to a differential equation with its boundary conditions, the form of that integral equation depends on the boundary conditions.

A second feature of integral equations is that their compact and completely self-contained form may turn out to be a more convenient or powerful formulation of a problem than a differential equation and its boundary conditions. Mathematical problems such as existence, uniqueness, and completeness may often be handled more easily and elegantly in integral form. And finally, whether or not we like it, there are problems, such as some diffusion and transport phenomena, that cannot be represented by differential equations. If we wish to solve such problems, we are forced to handle integral equations.

### Example 21.1.1 MOMENTUM REPRESENTATION IN QUANTUM MECHANICS

The Schrödinger equation (in ordinary space representation) for a particle of mass  $m$  subject to a potential  $V(\mathbf{r})$  is

$$-\frac{\hbar^2}{2m}\nabla^2\psi(\mathbf{r}) + V(\mathbf{r})\psi(\mathbf{r}) = E\psi(\mathbf{r}), \quad (21.5)$$

and we previously found, extending the 1-D result from Eq. (20.97), that in momentum space the equivalent equation (for the Coulomb potential in hartree atomic units) is

$$\frac{\mathbf{k}^2}{2m}\varphi(\mathbf{k}) + \frac{1}{(2\pi)^{3/2}} \int \frac{4\pi}{|\mathbf{k} - \mathbf{k}'|^2} \varphi(\mathbf{k}') d^3k' = E\varphi(\mathbf{k}). \quad (21.6)$$

This is an integral-equation eigenvalue problem. Note that the kernel of Eq. (21.6) is a function of  $\mathbf{k} - \mathbf{k}'$ ; this functional dependence, which arises from the convolution theorem,

is typical of an ordinary potential in which the direct-space wave function is multiplied by a function that depends only on position. ■

## Transformation of a Differential Equation into an Integral Equation

Often we find that we have a choice. The physical problem may be represented by a differential or an integral equation. Let us assume that we have the differential equation and wish to transform it into an integral equation. Starting with a **linear** second-order ordinary differential equation (ODE),

$$y'' + A(x)y' + B(x)y = g(x), \quad (21.7)$$

with initial conditions

$$y(a) = y_0, \quad y'(a) = y'_0,$$

we integrate to obtain

$$y'(x) = - \int_a^x A(t)y'(t) dt - \int_a^x B(t)y(t) dt + \int_a^x g(t) dt + y'_0.$$

Integrating the first integral on the right by parts yields

$$y'(x) = -A(x)y(x) - \int_a^x [B(t) - A'(t)]y(t) dt + \int_a^x g(t) dt + A(a)y_0 + y'_0.$$

Integrating a second time, we obtain

$$\begin{aligned} y(x) = & - \int_a^x A(t)y(t) dt - \int_a^x du \int_a^u [B(t) - A'(t)]y(t) dt \\ & + \int_a^x du \int_a^u g(t) dt + [A(a)y_0 + y'_0](x - a) + y_0. \end{aligned} \quad (21.8)$$

To transform this equation into a neater form, we use the relation

$$\int_a^x du \int_a^u f(t) dt = \int_a^x f(t) dt \int_t^x du = \int_a^x (x - t)f(t) dt. \quad (21.9)$$

Applying this result to Eq. (21.8), we obtain

$$\begin{aligned} y(x) = & - \int_a^x \left( A(t) + (x-t) \left[ B(t) - A'(t) \right] \right) y(t) dt \\ & + \int_a^x (x-t)g(t) dt + \left[ A(a)y_0 + y'_0 \right] (x-a) + y_0. \end{aligned} \quad (21.10)$$

If we now introduce the abbreviations

$$\begin{aligned} K(x, t) &= (t-x) \left[ B(t) - A'(t) \right] - A(t), \\ f(x) &= \int_a^x (x-t)g(t) dt + \left[ A(a)y_0 + y'_0 \right] (x-a) + y_0, \end{aligned}$$

Eq. (21.10) becomes

$$y(x) = f(x) + \int_a^x K(x, t)y(t) dt, \quad (21.11)$$

which is a Volterra equation of the second kind. Note that  $f(x)$  in Eq. (21.11) has a form that includes the initial conditions from the original differential equation.

Another method for obtaining an integral equation equivalent to a differential equation plus its boundary conditions was presented in Section 10.1, where we found that the Green's function for a differential equation appeared as the kernel of the equivalent integral equation.

### Example 21.1.2 LINEAR OSCILLATOR EQUATION

Let's find an integral equation equivalent to the linear oscillator equation

$$y'' + \omega^2 y = 0 \quad (21.12)$$

with boundary conditions

$$y(0) = 0, \quad y'(0) = 1.$$

This corresponds to Eq. (21.7) with

$$A(x) = 0, \quad B(x) = \omega^2, \quad g(x) = 0.$$

Substituting into Eq. (21.10), we find that the integral equation becomes

$$y(x) = x + \omega^2 \int_0^x (t-x)y(t) dt. \quad (21.13)$$

This integral equation, Eq. (21.13), is equivalent to the original differential equation plus the initial conditions. A check shows that each form is indeed satisfied by  $y(x) = (1/\omega) \sin \omega x$ .

Let us reconsider the linear oscillator equation, Eq. (21.12), but now with the boundary conditions

$$y(0) = 0, \quad y(b) = 0.$$

Since  $y'(0)$  is not given, we must modify the procedure. The first integration gives

$$y' = -\omega^2 \int_0^x y \, dx + y'(0).$$

Integrating a second time and again using Eq. (21.9), we have

$$y = -\omega^2 \int_0^x (x-t)y(t) \, dt + xy'(0). \quad (21.14)$$

To eliminate the unknown  $y'(0)$ , we now impose the condition  $y(b) = 0$ . This gives

$$\omega^2 \int_0^b (b-t)y(t) \, dt = by'(0).$$

Substituting this back into Eq. (21.14), we obtain

$$y(x) = -\omega^2 \int_0^x (x-t)y(t) \, dt + \omega^2 \frac{x}{b} \int_0^b (b-t)y(t) \, dt.$$

Now let us break the interval  $[0, b]$  into two intervals,  $[0, x]$  and  $[x, b]$ . Since

$$\frac{x}{b}(b-t) - (x-t) = \frac{t}{b}(b-x),$$

we find

$$y(x) = \omega^2 \int_0^x \frac{t}{b}(b-x)y(t) \, dt + \omega^2 \int_x^b \frac{x}{b}(b-t)y(t) \, dt. \quad (21.15)$$

Finally, if we define the kernel

$$K(x, t) = \begin{cases} \frac{t}{b}(b-x), & t < x, \\ \frac{x}{b}(b-t), & t > x, \end{cases} \quad (21.16)$$

we have

$$y(x) = \omega^2 \int_0^b K(x, t) y(t) dt, \quad (21.17)$$

a homogeneous Fredholm equation of the second kind.

Our new kernel,  $K(x, t)$ , illustrated in Fig. 21.1, has some interesting properties.

1. It is symmetric,  $K(x, t) = K(t, x)$ .
2. It is continuous, in the sense that

$$\left. \frac{t}{b}(b-x) \right|_{t=x} = \left. \frac{x}{b}(b-t) \right|_{t=x}.$$

3. Its derivative with respect to  $t$  is **discontinuous**. As  $t$  increases through the point  $t = x$ , there is a discontinuity of  $-1$  in  $\partial K(x, t)/\partial t$ .

Comparing with the discussion in Section 10.1, we identify  $K(x, t)$  as the Green's function for this ODE **with the specified boundary conditions**. Note in particular Eq. (10.30), which corresponds exactly to what was found here. ■

The above example shows how the initial or boundary conditions play a decisive role in the conversion of a linear second-order ODE into an integral equation. Summarizing,

*If we have **initial** conditions (only one end of our interval), the differential equation transforms into a Volterra integral equation. But if we have a **boundary value problem** (boundary conditions at both ends of our interval), the differential equation leads to a Fredholm-type integral equation with a kernel that will be the Green's function appropriate to the given boundary conditions.*

In closing, we call attention to the fact that the reverse transformation (integral equation to differential equation) is not always possible. There exist integral equations for which no corresponding differential equation is known.

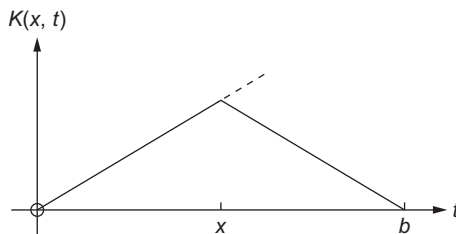


FIGURE 21.1 Kernel, Eq. (21.16), for linear oscillator boundary-value problem.

## Exercises

**21.1.1** Starting with the ODE, integrate twice and derive the Volterra integral equation corresponding to

(a)  $y''(x) - y(x) = 0; \quad y(0) = 0, \quad y'(0) = 1.$

*ANS.*  $y = \int_0^x (x-t)y(t) dt + x.$

(b)  $y''(x) - y(x) = 0; \quad y(0) = 1, \quad y'(0) = -1.$

*ANS.*  $y = \int_0^x (x-t)y(t) dt - x + 1.$

Check your results with [Eq. \(21.11\)](#).

**21.1.2** Starting with the given answers of [Exercise 21.1.1](#), differentiate and recover the original ODEs **and the boundary conditions**.

**21.1.3** Given  $\varphi(x) = x - \int_0^x (t-x)\varphi(t) dt,$

solve this integral equation by converting it to an ODE (plus boundary conditions) and solving the ODE (by inspection).

**21.1.4** Show that the homogeneous Volterra equation of the second kind

$$\psi(x) = \lambda \int_0^x K(x, t)\psi(t) dt$$

has no solution (apart from the trivial solution  $\psi = 0$ ).

*Hint.* Develop a Maclaurin expansion of  $\psi(x)$ . Assume that  $\psi(x)$  and  $K(x, t)$  are differentiable with respect to  $x$  as needed.

## 21.2 SOME SPECIAL METHODS

It is well known that general methods are available both for differentiating functions and (compare Chapters 7 and 9) for solving linear differential equations, while there is no general direct method for evaluating integrals. Integrations are carried out using a variety of tools of limited applicability, and the process is ultimately one of pattern recognition and the application of experience. Similar observations apply to the solution of integral equations. We consider here some special methods that work when the integral equation under study has suitable characteristics.

## Integral-Transform Methods

When the kernel of an integral equation (and its integration limits) match the specification of an integral transform for which we have an inversion formula, we can use that identification to solve the integral equation. Formulas based on four integral transforms are listed here for reference, in each case with  $f(x)$  a known function and  $\varphi(x)$  to be determined.

If our integral equation is  $f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{ixt} \varphi(t) dt$ , then its solution is

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-ixt} f(t) dt \quad (\text{Fourier transform}). \quad (21.18)$$

If our integral equation is  $f(x) = \int_0^{\infty} e^{-xt} \varphi(t) dt$ , then its solution is

$$\varphi(x) = \frac{1}{2\pi i} \int_{\gamma-i\infty}^{\gamma+i\infty} e^{xt} f(t) dt \quad (\text{Laplace transform}). \quad (21.19)$$

If our integral equation is  $f(x) = \int_0^{\infty} t^{x-1} \varphi(t) dt$ , then its solution is

$$\varphi(x) = \frac{1}{2\pi i} \int_{\gamma-i\infty}^{\gamma+i\infty} x^{-t} f(t) dt \quad (\text{Mellin transform}). \quad (21.20)$$

If our integral equation is  $f(x) = \int_0^{\infty} t \varphi(t) J_{\nu}(xt) dt$ , then its solution is

$$\varphi(x) = \int_0^{\infty} t f(t) J_{\nu}(xt) dt \quad (\text{Hankel transform}). \quad (21.21)$$

Note that these formulas can also be applied “in reverse,” i.e., with  $\varphi(x)$  known and  $f(x)$  to be determined. This observation, however, is of somewhat limited utility since nothing significantly new appears for the inverse Fourier and Hankel transforms, while the integration limits for the inverse Laplace and Mellin transforms make them unlikely to appear in an integral equation.

Actually the usefulness of the integral transform technique extends a bit beyond these four rather specialized forms. We illustrate with two examples.

### Example 21.2.1 FOURIER TRANSFORM SOLUTION

Let's consider a Fredholm equation of the first kind with a kernel of the general type  $k(x-t)$ , where  $k$  is a function (not a constant),

$$f(x) = \int_{-\infty}^{\infty} k(x-t) \varphi(t) dt, \quad (21.22)$$



in which  $\varphi(t)$  is our unknown function. **Assuming that the needed transforms exist**, we apply the Fourier convolution theorem, [Eq. \(20.71\)](#), to obtain

$$f(x) = \int_{-\infty}^{\infty} K(\omega)\Phi(\omega)e^{-i\omega x}d\omega. \quad (21.23)$$

The functions  $K(\omega)$  and  $\Phi(\omega)$  are, respectively, the Fourier transforms of  $k(x)$  and  $\varphi(x)$ . Taking the Fourier transform of both sides of [Eq. \(21.23\)](#), the formula for which is [Eq. \(21.18\)](#), we find

$$K(\omega)\Phi(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} f(x)e^{i\omega x}dx = \frac{F(\omega)}{\sqrt{2\pi}}, \quad (21.24)$$

where  $F(\omega)$  is the Fourier transform of  $f(x)$ . Since  $\Phi(\omega)$  is the only unknown in [Eq. \(21.24\)](#), we may solve for it, obtaining

$$\Phi(\omega) = \frac{1}{\sqrt{2\pi}} \frac{F(\omega)}{K(\omega)}, \quad (21.25)$$

and, using the inverse Fourier transform, we have the solution to [Eq. \(21.22\)](#):

$$\varphi(x) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{F(\omega)}{K(\omega)} e^{-i\omega x} d\omega. \quad (21.26)$$

A rigorous justification of this result is presented by Morse and Feshbach (see Additional Readings). An extension of this transformation solution appears as [Exercise 21.2.1](#). ■

### **Example 21.2.2** GENERALIZED ABEL EQUATION

The generalized Abel equation is a Volterra equation of the first kind:

$$f(x) = \int_0^x \frac{\varphi(t)}{(x-t)^\alpha} dt, \quad 0 < \alpha < 1, \quad \text{with } \begin{cases} f(x) & \text{known,} \\ \varphi(t) & \text{unknown.} \end{cases} \quad (21.27)$$

Taking the Laplace transform of both sides of this equation, we obtain

$$\mathcal{L}\{f(x)\} = \mathcal{L}\left\{\int_0^x \frac{\varphi(t)}{(x-t)^\alpha} dt\right\} = \mathcal{L}\{x^{-\alpha}\}\mathcal{L}\{\varphi(x)\},$$

the last step following by the Laplace convolution theorem, Eq. (20.186). Then, evaluating  $\mathcal{L}\{x^{-\alpha}\}$  from entry 3 of Table 20.1,

$$\mathcal{L}\{\varphi(x)\} = \frac{s^{1-\alpha} \mathcal{L}\{f(x)\}}{\Gamma(1-\alpha)}. \quad (21.28)$$

In principle, our integral equation is solved, since all that remains is to take the inverse transform of Eq. (21.28). A clever way of obtaining the inverse transform proceeds as follows, with its initial step being to divide Eq. (21.28) by  $s$ .<sup>1</sup> We get

$$\frac{1}{s} \mathcal{L}\{\varphi(x)\} = \frac{s^{-\alpha} \mathcal{L}\{f(x)\}}{\Gamma(1-\alpha)} = \frac{\mathcal{L}\{x^{\alpha-1}\} \mathcal{L}\{f(x)\}}{\Gamma(\alpha) \Gamma(1-\alpha)}.$$

Combining the gamma functions according to Eq. (13.23) and applying the Laplace convolution theorem again, we discover that

$$\frac{1}{s} \mathcal{L}\{\varphi(x)\} = \frac{\sin \pi \alpha}{\pi} \mathcal{L} \left\{ \int_0^x \frac{f(t)}{(x-t)^{1-\alpha}} dt \right\}.$$

Inverting with the aid of Entry 3 of Table 20.2, we get

$$\int_0^x \varphi(t) dt = \frac{\sin \pi \alpha}{\pi} \int_0^x \frac{f(t)}{(x-t)^{1-\alpha}} dt,$$

and finally, by differentiating, we have the solution to our generalized Abel equation:

$$\varphi(x) = \frac{\sin \pi \alpha}{\pi} \frac{d}{dx} \int_0^x \frac{f(t)}{(x-t)^{1-\alpha}} dt. \quad (21.29)$$

■

## Generating-Function Method

Occasionally, the reader may encounter integral equations that involve generating functions. Suppose we have the admittedly special case,

$$f(x) = \int_{-1}^1 \frac{\varphi(t)}{(1-2xt+x^2)^{1/2}} dt, \quad -1 \leq x \leq 1, \quad (21.30)$$

where  $f(x)$  is known and  $\varphi(t)$  is to be determined.

We note two important features:

1.  $(1-2xt+x^2)^{-1/2}$  generates the Legendre polynomials.
2.  $[-1, 1]$  is the orthogonality interval for the Legendre polynomials.

<sup>1</sup>This division converts  $s^{1-\alpha}$ , which cannot be inverted when  $0 < \alpha < 1$ , into  $s^{-\alpha}$ , which is the transform of  $x^{\alpha-1}/\Gamma(\alpha)$ .

These features make it possible to expand the denominator in Legendre polynomials, suggesting that it may be useful also to represent  $\varphi(t)$  as an expansion in these same functions. Thus, we introduce the expansions

$$\frac{1}{(1-2xt+x^2)^{1/2}} = \sum_{n=0}^{\infty} P_n(t)x^n, \quad \varphi(t) = \sum_{m=0}^{\infty} a_m P_m(t).$$

Substituting these expansions into our integral equation, Eq. (21.30),

$$\begin{aligned} f(x) &= \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} a_m x^n \int_{-1}^1 P_n(t) P_m(t) dt = \sum_{n=0}^{\infty} \sum_{m=0}^{\infty} a_m x^n \frac{2\delta_{nm}}{2n+1} \\ &= \sum_{n=0}^{\infty} \frac{2a_n}{2n+1} x^n. \end{aligned} \quad (21.31)$$

If we now insert into Eq. (21.31) the Maclaurin series expansion for  $f(x)$ ,

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n,$$

we may equate powers of  $x$ , reaching, for each  $n$ ,

$$\frac{f^{(n)}(0)}{n!} = \frac{2a_n}{2n+1},$$

so the solution to our integral equation is

$$\varphi(t) = \sum_{n=0}^{\infty} \frac{2n+1}{2} \frac{f^{(n)}(0)}{n!} P_n(t). \quad (21.32)$$

Similar results may be obtained with other generating functions (see the list in Table 12.1).

*This technique of expanding in a series of special functions is always available. It is worth a try whenever the expansion is possible (and convenient) and the interval is appropriate.*

## Separable Kernel

We consider here the special case that the kernel of our integral equation is separable, in the sense that

$$K(x, t) = \sum_{j=1}^n M_j(x) N_j(t), \quad (21.33)$$

where  $n$ , the upper limit of the sum, is **finite**. Such kernels are sometimes called **degenerate**. Our class of separable kernels includes all polynomials and many of the elementary transcendental functions. For example,  $K(x, t) = \cos(t - x)$  is separable:

$$\cos(t - x) = \cos t \cos x + \sin t \sin x.$$

Integral equations with separable kernels have the desirable property that they can be related to eigenvalue equations and permit the application of methods of linear algebra.

Let's consider a Fredholm equation of the second kind, Eq. (21.2), with a separable kernel of the form given in Eq. (21.33). Inserting this formula for  $K(x, t)$  and bringing the summation outside the integral, we have

$$\varphi(x) = f(x) + \lambda \sum_{j=1}^n M_j(x) \int_a^b N_j(t) \varphi(t) dt. \quad (21.34)$$

We now see that the integral with respect to  $t$  will for each  $j$  be a constant (with values that are currently not known):

$$\int_a^b N_j(t) \varphi(t) dt = c_j. \quad (21.35)$$

Hence Eq. (16.71) becomes

$$\varphi(x) = f(x) + \lambda \sum_{j=1}^n c_j M_j(x). \quad (21.36)$$

Once the constants  $c_i$  have been determined, Eq. (21.36) will give us  $\varphi(x)$ , the solution to our integral equation. Equation (21.36) further tells us that the form of  $\varphi(x)$  will consist of  $f(x)$  plus a linear combination of the  $x$ -dependent factors in the separable kernel.

We may find the  $c_i$  by multiplying Eq. (21.36) by  $N_i(x)$  and integrating to eliminate the  $x$ -dependence. Use of Eq. (21.35) yields

$$c_i = b_i + \lambda \sum_{j=1}^n a_{ij} c_j, \quad (21.37)$$

where

$$b_i = \int_a^b N_i(x) f(x) dx, \quad a_{ij} = \int_a^b N_i(x) M_j(x) dx. \quad (21.38)$$

It is perhaps helpful to write Eq. (21.37) in matrix form, with  $\mathbf{A} = (a_{ij})$ :

$$\mathbf{b} = \mathbf{c} - \lambda \mathbf{A} \mathbf{c} = (1 - \lambda \mathbf{A}) \mathbf{c}, \quad (21.39)$$

or

$$\mathbf{c} = (1 - \lambda \mathbf{A})^{-1} \mathbf{b}. \quad (21.40)$$

Equation (21.39) is equivalent to a set of simultaneous linear algebraic equations

$$\begin{aligned} (1 - \lambda a_{11})c_1 - \lambda a_{12}c_2 - \lambda a_{13}c_3 - \cdots &= b_1, \\ -\lambda a_{21}c_1 + (1 - \lambda a_{22})c_2 - \lambda a_{23}c_3 - \cdots &= b_2, \\ -\lambda a_{31}c_1 - \lambda a_{32}c_2 + (1 - \lambda a_{33})c_3 - \cdots &= b_3, \quad \text{and so on.} \end{aligned} \quad (21.41)$$

If our integral equation is homogeneous, so  $f(x) = 0$ , then  $\mathbf{b} = 0$ . To get a solution in that case, we set the determinant of the coefficients of  $c_i$  equal to zero:

$$|1 - \lambda \mathbf{A}| = 0, \quad (21.42)$$

exactly as for any matrix eigenvalue problem. The roots of Eq. (21.42) yield our eigenvalues. Substituting into  $(1 - \lambda \mathbf{A})\mathbf{c} = 0$ , we find the  $c_i$  and then Eq. (21.36) gives our solution.

### Example 21.2.3 HOMOGENEOUS FREDHOLM EQUATION

To illustrate this technique for determining eigenvalues and eigenfunctions of the homogeneous Fredholm equation of the second kind, we consider

$$\varphi(x) = \lambda \int_{-1}^1 (t+x)\varphi(t) dt. \quad (21.43)$$

Writing the kernel of this equation as  $M_1(x)N_1(t) + M_2(x)N_2(t)$ , we have

$$\begin{aligned} M_1(x) &= 1, & M_2(x) &= x, \\ N_1(t) &= t, & N_2(t) &= 1. \end{aligned}$$

Using the notation of Eqs. (21.33) to (21.42), we find from Eq. (21.38):

$$a_{11} = a_{22} = 0, \quad a_{12} = \frac{2}{3}, \quad a_{21} = 2; \quad b_1 = b_2 = 0.$$

Equation (21.42), our secular equation, becomes<sup>2</sup>

$$\begin{vmatrix} 1 - \frac{2\lambda}{3} & 2\lambda \\ -2\lambda & 1 \end{vmatrix} = 0. \quad (21.44)$$

Expanding, we obtain

$$1 - \frac{4\lambda^2}{3} = 0, \quad \lambda = \pm \frac{\sqrt{3}}{2}. \quad (21.45)$$

Substituting the eigenvalues  $\lambda = \pm \sqrt{3}/2$  into Eq. (21.39), we have

$$c_1 \mp \frac{c_2}{\sqrt{3}} = 0. \quad (21.46)$$

Finally, with the choice  $c_1 = 1$ , Eq. (21.36) gives the two solutions

$$\varphi_1(x) = \frac{\sqrt{3}}{2}(1 + \sqrt{3}x), \quad \lambda = \frac{\sqrt{3}}{2}, \quad (21.47)$$

<sup>2</sup>This equation would look more like our usual secular equations if each row of the determinant were divided by  $\lambda$ . Then we would have the secular equation in a familiar form, but with  $1/\lambda$  identified as the eigenvalue.

$$\varphi_2(x) = -\frac{\sqrt{3}}{2}(1 - \sqrt{3}x), \quad \lambda = -\frac{\sqrt{3}}{2}. \quad (21.48)$$

Since our equation is homogeneous, the normalization of  $\varphi(x)$  is arbitrary. ■

If the kernel of an integral equation is not separable in the sense of Eq. (21.33), there is still the possibility that it may be approximated by a kernel that is separable. Then we can get the exact solution of an approximate equation, which we can treat as an approximation to the solution of the original equation.

## Exercises

**21.2.1** The kernel of a Fredholm equation of the second kind,

$$\varphi(x) = f(x) + \lambda \int_{-\infty}^{\infty} K(x, t)\varphi(t) dt,$$

is of the form  $k(x - t)$ .<sup>3</sup> Assuming that the required transforms exist, show that

$$\varphi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \frac{F(t)e^{-ixt} dt}{1 - \sqrt{2\pi}\lambda K(t)}.$$

$F(t)$  and  $K(t)$  are the Fourier transforms of  $f(x)$  and  $k(x)$ , respectively.

**21.2.2** (a) The kernel of a Volterra equation of the first kind,

$$f(x) = \int_0^x K(x, t)\varphi(t) dt,$$

has the form  $k(x - t)$ . Assuming that the required transforms exist, show that

$$\varphi(x) = \frac{1}{2\pi i} \int_{\gamma-i\infty}^{\gamma+i\infty} \frac{F(s)}{K(s)} e^{xs} ds,$$

where  $F(s)$  and  $K(s)$  are, respectively, the Laplace transforms of  $f(x)$  and  $k(x)$ .

(b) In terms of the notation of part (a), show that the Volterra equation of the second kind,

$$\varphi(x) = f(x) + \lambda \int_0^x K(x, t)\varphi(t) dt,$$

<sup>3</sup>This kernel and a range  $0 \leq x < \infty$  are the characteristics of integral equations of the Wiener-Hopf type. Details will be found in Chapter 8 of Morse and Feshbach (1953); see the Additional Readings.

has solution

$$\varphi(x) = \frac{1}{2\pi i} \int_{\gamma-i\infty}^{\gamma+i\infty} \frac{F(s)}{1-\lambda K(s)} e^{xs} ds.$$

**21.2.3** Using the Laplace transform solution ([Exercise 21.2.2](#)), solve

$$(a) \quad \varphi(x) = x + \int_0^x (t-x)\varphi(t) dt.$$

*ANS.*  $\varphi(x) = \sin x$ .

$$(b) \quad \varphi(x) = x - \int_0^x (t-x)\varphi(t) dt.$$

*ANS.*  $\varphi(x) = \sinh x$ .

Check your results by substituting back into the original integral equations.

**21.2.4** Reformulate the equations of [Example 21.2.1](#) for integrals on the range  $(0, \infty)$  using Fourier cosine transforms.

**21.2.5** Given the Fredholm integral equation,

$$e^{-x^2} = \int_{-\infty}^{\infty} e^{-(x-t)^2} \varphi(t) dt,$$

apply the Fourier convolution technique of [Example 21.2.1](#) to solve for  $\varphi(t)$ .

**21.2.6** Solve Abel's equation,

$$f(x) = \int_0^x \frac{\varphi(t)}{(x-t)^\alpha} dt, \quad 0 < \alpha < 1,$$

by the following method:

- Multiply both sides by  $(z-x)^{\alpha-1}$  and integrate with respect to  $x$  over the range  $0 \leq x \leq z$ .
- Reverse the order of integration and evaluate the integral on the right-hand side (with respect to  $x$ ) by recognizing it as a beta function.

*Note.*

$$\int_t^z \frac{dx}{(z-x)^{1-\alpha}(x-t)^\alpha} = B(1-\alpha, \alpha) = \Gamma(\alpha)\Gamma(1-\alpha) = \frac{\pi}{\sin \pi \alpha}.$$

**21.2.7** Given the generalized Abel equation with  $f(x) = 1$ ,

$$1 = \int_0^x \frac{\varphi(t)}{(x-t)^\alpha} dt, \quad 0 < \alpha < 1,$$

solve for  $\varphi(t)$  and verify that  $\varphi(t)$  is a solution of the given equation.

$$ANS. \quad \varphi(t) = \frac{\sin \pi \alpha}{\pi} t^{\alpha-1}.$$

**21.2.8** A Fredholm equation of the first kind has a kernel  $e^{-(x-t)^2}$ :

$$f(x) = \int_{-\infty}^{\infty} e^{-(x-t)^2} \varphi(t) dt.$$

Show that the solution is

$$\varphi(x) = \frac{1}{\sqrt{\pi}} \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{2^n n!} H_n(x),$$

in which  $H_n(x)$  is an  $n$ th-order Hermite polynomial.

**21.2.9** Solve the integral equation

$$f(x) = \int_{-1}^1 \frac{\varphi(t)}{(1-2xt+x^2)^{1/2}} dt, \quad -1 \leq x \leq 1,$$

for the unknown function  $\varphi(t)$ , if

(a)  $f(x) = x^{2s}$ ,

(b)  $f(x) = x^{2s+1}$ .

$$ANS. \quad (a) \varphi(t) = \frac{4s+1}{2} P_{2s}(t), \quad (b) \varphi(t) = \frac{4s+3}{2} P_{2s+1}(t).$$

**21.2.10** Find the eigenvalues and eigenfunctions of

$$\varphi(x) = \lambda \int_{-1}^1 (t-x) \varphi(t) dt.$$

**21.2.11** Find the eigenvalues and eigenfunctions of

$$\varphi(x) = \lambda \int_0^{2\pi} \cos(x-t) \varphi(t) dt.$$

$$ANS. \quad \lambda_1 = \lambda_2 = \frac{1}{\pi}, \quad \varphi(x) = A \cos x + B \sin x.$$



**21.2.12** Find the eigenvalues and eigenfunctions of

$$y(x) = \lambda \int_{-1}^1 (x-t)^2 y(t) dt.$$

*Hint.* This problem may be treated by the separable-kernel method or by a Legendre expansion.

**21.2.13** Use the separable-kernel technique to show that

$$\psi(x) = \lambda \int_0^\pi \cos x \sin t \psi(t) dt$$

has **no** solution (apart from  $\psi = 0$ ). Explain this result in terms of separability and symmetry.

**21.2.14** Given  $\varphi(x) = \lambda \int_0^1 (1+xt)\varphi(t) dt$ ,

solve for the eigenvalues and the eigenfunctions by the separable-kernel technique.

**21.2.15** Knowing the form of the solutions of an integral equation can be a great advantage. For

$$\varphi(x) = \lambda \int_0^1 (1+xt)\varphi(t) dt,$$

assume  $\varphi(x)$  to have the form  $1+bx$ . Substitute into the integral equation. Integrate and solve for  $b$  and  $\lambda$ .

**21.2.16** The equation

$$f(x) = \int_a^b K(x, t)\varphi(t) dt$$

has a degenerate kernel  $K(x, t) = \sum_{i=1}^n M_i(x)N_i(t)$ .

(a) Show that this integral equation has no solution unless  $f(x)$  can be written as

$$f(x) = \sum_{i=1}^n f_i M_i(x),$$

where the  $f_i$  are constants.

(b) Show that to any solution  $\varphi(x)$  we may add  $\psi(x)$ , provided that  $\psi(x)$  is orthogonal to all  $N_i(x)$ :

$$\int_a^b N_i(x)\psi(x)dx = 0 \quad \text{for all } i.$$

**21.2.17** A Kirchhoff diffraction theory analysis of a laser leads to the integral equation

$$v(\mathbf{r}_2) = \gamma \iint K(\mathbf{r}_1, \mathbf{r}_2) v(\mathbf{r}_1) dA.$$

The unknown,  $v(\mathbf{r}_1)$ , gives the geometric distribution of the radiation field over one mirror surface; the range of integration is over the surface of that mirror. For square confocal spherical mirrors, the integral equation becomes

$$v(x_2, y_2) = \frac{-i\gamma e^{ikb}}{\lambda b} \int_{-a}^a \int_{-a}^a e^{-(ik/b)(x_1 x_2 + y_1 y_2)} v(x_1, y_1) dx_1 dy_1,$$

in which  $b$  is the centerline distance between the laser mirrors. This can be put in a somewhat simpler form by the substitutions

$$\frac{kx_i^2}{b} = \xi_i^2, \quad \frac{ky_i^2}{b} = \eta_i^2, \quad \text{and} \quad \frac{ka^2}{b} = \frac{2\pi a^2}{\lambda b} = \alpha^2.$$

- (a) Show that the variables separate and we get two integral equations.
- (b) Show that the new limits,  $\pm\alpha$ , may be approximated by  $\pm\infty$  for a mirror dimension  $a \gg \lambda$ .
- (c) Solve the resulting integral equations.

## 21.3 NEUMANN SERIES

Many and probably most integral equations cannot be solved by the specialized techniques of the preceding section. Here we develop a rather general technique for solving integral equations. The method, due largely to Neumann, Liouville, and Volterra, develops the unknown function  $\varphi(x)$  as a power series in  $\lambda$ , where  $\lambda$  is a given constant. The method is applicable whenever the series converges.

We solve a linear integral equation of the second kind by successive approximations; let's take as an example the Fredholm equation

$$\varphi(x) = f(x) + \lambda \int_a^b K(x, t) \varphi(t) dt, \quad (21.49)$$

in which  $f(x) \neq 0$ . If the upper limit of the integral is a variable (Volterra equation), the following development will still hold, but with minor modifications. Let us make the following initial approximation to our unknown function:

$$\varphi(x) \approx \varphi_0(x) = f(x). \quad (21.50)$$

This choice is not mandatory. If you can make a better guess, go ahead and guess. The choice here is equivalent to saying that the term of the equation containing the integral is

small relative to  $f(x)$ . To improve this first crude approximation, we feed  $\varphi_0(x)$  back into the integral in Eq. (21.49), getting

$$\varphi_1(x) = f(x) + \lambda \int_a^b K(x, t) f(t) dt. \quad (21.51)$$

Substituting the new  $\varphi_1(x)$  back into Eq. (21.49), we obtain a second approximation to  $\varphi(x)$ :

$$\begin{aligned} \varphi_2(x) = f(x) + \lambda \int_a^b K(x, t_1) f(t_1) dt_1 \\ + \lambda^2 \int_a^b \int_a^b K(x, t_1) K(t_1, t_2) f(t_2) dt_2 dt_1. \end{aligned}$$

This process can be repeated indefinitely, defining after  $n$  steps the  $n$ th order approximation

$$\varphi_n(x) = \sum_{i=0}^n \lambda^i u_i(x), \quad (21.52)$$

where

$$\begin{aligned} u_0(x) &= f(x) \\ u_1(x) &= \int_a^b K(x, t_1) f(t_1) dt_1 \\ u_2(x) &= \int_a^b \int_a^b K(x, t_1) K(t_1, t_2) f(t_2) dt_2 dt_1 \\ u_n(x) &= \int_a^b \int_a^b \cdots \int_a^b K(x, t_1) K(t_1, t_2) \cdots K(t_{n-1}, t_n) f(t_n) dt_n \cdots dt_1. \end{aligned} \quad (21.53)$$

We expect that our solution  $\varphi(x)$  will be

$$\varphi(x) = \lim_{n \rightarrow \infty} \varphi_n(x) = \lim_{n \rightarrow \infty} \sum_{i=0}^n \lambda^i u_i(x), \quad (21.54)$$

**provided that our infinite series converges.**

We may conveniently check the convergence by the Cauchy ratio test, Section 1.1, noting that

$$|\lambda^n u_n(x)| \leq |\lambda^n| \cdot |f|_{\max} \cdot |K|_{\max}^n \cdot |b - a|^n,$$

using  $|f|_{\max}$  to represent the **maximum** value of  $|f(x)|$  in the interval  $[a, b]$  and  $|K|_{\max}$  to represent the maximum value of  $|K(x, t)|$  in its domain in the  $xt$ -plane. A sufficient condition for convergence is

$$|\lambda| \cdot |K|_{\max} \cdot |b - a| < 1. \quad (21.55)$$

Note that  $\lambda|u_n(\max)|$  is being used as a **comparison** series. If it converges, our actual series must converge. If this condition is not satisfied, we may or may not have convergence, and a more sensitive test would be required to determine the convergence. Of course, even if the Neumann series diverges, there still may be a solution to our integral equation obtainable by another method.

To gain more understanding of our iterative manipulation, we may find it helpful to rewrite the Neumann series solution, Eq. (21.54), in operator form. We start by rewriting Eq. (21.49) as

$$\varphi = \lambda K \varphi + f,$$

where  $K$  represents the **integral operator**  $\int_a^b K(x, t)[\ ] dt$ . Solving symbolically for  $\varphi$ , we obtain

$$\varphi = (1 - \lambda K)^{-1} f.$$

Binomial expansion leads to Eq. (21.54). The **convergence** of the Neumann series is a demonstration that the inverse operator  $(1 - \lambda K)^{-1}$  exists.

### Example 21.3.1 NEUMANN SERIES SOLUTION

To illustrate the Neumann method, we consider the integral equation

$$\varphi(x) = x + \frac{1}{2} \int_{-1}^1 (t - x) \varphi(t) dt. \quad (21.56)$$

To start the Neumann series, we take

$$\varphi_0(x) = x.$$

Then

$$\varphi_1(x) = x + \frac{1}{2} \int_{-1}^1 (t - x) t dt = x + \frac{1}{2} \left( \frac{1}{3} t^3 - \frac{1}{2} t^2 x \right) \Big|_{-1}^1 = x + \frac{1}{3}.$$

Substituting  $\varphi_1(x)$  back into Eq. (21.56), we get

$$\varphi_2(x) = x + \frac{1}{2} \int_{-1}^1 (t - x) t dt + \frac{1}{2} \int_{-1}^1 (t - x) \frac{1}{3} dt = x + \frac{1}{3} - \frac{x}{3}.$$

Continuing this process of substituting back into Eq. (21.56), we obtain

$$\varphi_3(x) = x + \frac{1}{3} - \frac{x}{3} - \frac{1}{3^2},$$

and by mathematical induction (Section 1.4),

$$\varphi_{2n}(x) = x + \sum_{s=1}^n (-1)^{s-1} 3^{-s} - x \sum_{s=1}^n (-1)^{s-1} 3^{-s}. \quad (21.57)$$

Letting  $n \rightarrow \infty$ , we get

$$\varphi(x) = \frac{3}{4}x + \frac{1}{4}. \quad (21.58)$$

This solution can (and should) be checked by substituting back into the original equation, Eq. (21.56).

It is interesting to note that our series converged easily even though Eq. (21.55) is **not** satisfied in this particular case. Actually Eq. (21.55) is a rather crude upper bound on  $\lambda$ . It can be shown that a necessary and sufficient condition for the convergence of our series solution is that  $|\lambda| < |\lambda_e|$ , where  $\lambda_e$  is the eigenvalue of smallest magnitude of the corresponding homogeneous equation (that with  $f(x) = 0$ ). For this particular example,  $\lambda_e = \sqrt{3}/2$ . Clearly,  $\lambda = \frac{1}{2} < \lambda_e$ . ■

The technique illustrated by the Neumann series occurs in a number of contexts in quantum mechanics. For example, one approach to the calculation of time-dependent perturbations in quantum mechanics starts with the integral equation for the evolution operator

$$U(t, t_0) = 1 - \frac{i}{\hbar} \int_{t_0}^t dt_1 V(t_1) U(t_1, t_0). \quad (21.59)$$

Iteration leads to

$$U(t, t_0) = 1 - \frac{i}{\hbar} \int_{t_0}^t dt_1 V(t_1) + \left(\frac{i}{\hbar}\right)^2 \int_{t_0}^t dt_1 \int_{t_0}^{t_1} dt_2 V(t_1) V(t_2) + \cdots. \quad (21.60)$$

The evolution operator is obtained as a series of multiple integrals of the perturbing potential  $V(t)$ , closely analogous to the Neumann series, Eq. (21.52).

A second and similar relationship between the Neumann series and quantum mechanics appears when the Schrödinger wave equation for scattering is reformulated as an integral equation. See Example 10.2.2. The first term in a Neumann series solution is the incident (unperturbed) wave. The second term is the first-order Born approximation, Eq. (10.51).

The Neumann method may also be applied to Volterra integral equations of the second kind, corresponding to replacing the fixed upper limit  $b$  in Eq. (21.49) by a variable,  $x$ . In the Volterra case the Neumann series converges for all  $\lambda$  as long as the kernel is square integrable.

**Exercises****21.3.1** Using the Neumann series, solve

$$(a) \quad \varphi(x) = 1 - 2 \int_0^x t \varphi(t) dt,$$

ANS. (a)  $\varphi(x) = e^{-x^2}$ .

$$(b) \quad \varphi(x) = x + \int_0^x (t-x)\varphi(t) dt,$$

$$(c) \quad \varphi(x) = x - \int_0^x (t-x)\varphi(t) dt.$$

**21.3.2** Solve

$$\psi(x) = x + \int_0^1 (1+xt)\psi(t) dt$$

by each of the following methods:

- (a) The Neumann series technique,
- (b) The separable-kernel technique,
- (c) Educated guessing.

**21.3.3** Solve

$$\varphi(x) = 1 + \lambda^2 \int_0^x (x-t)\varphi(t) dt$$

by each of the following methods:

- (a) Reduction to an ODE (find the boundary conditions),
- (b) The Neumann series,
- (c) The use of Laplace transforms.

ANS.  $\varphi(x) = \cosh \lambda x$ .**21.3.4** (a) In Eq. (21.59), take  $V = V_0$ , independent of  $t$ . Without using Eq. (21.60), show that Eq. (21.59) leads directly to

$$U(t-t_0) = \exp \left[ -\frac{i}{\hbar} (t-t_0) V_0 \right].$$

- (b) Repeat for Eq. (21.60) without using Eq. (21.59).

## 21.4 HILBERT-SCHMIDT THEORY

### Symmetrization of Kernels

The Hilbert-Schmidt theory deals with linear integral equations of the Fredholm type with symmetric kernels:

$$K(x, t) = K(t, x). \quad (21.61)$$

The symmetry is of great importance, both because we will find it leads to results parallel to those found for the Sturm-Liouville theory of differential equations, and also because many problems of physical relevance can be written as Fredholm integral equations with symmetric kernels.

Before plunging into the theory, we note that some important nonsymmetric kernels can be symmetrized. If we have the equation

$$\varphi(x) = f(x) + \lambda \int_a^b K(x, t) \rho(t) \varphi(t) dt, \quad (21.62)$$

the total kernel is actually  $K(x, t) \rho(t)$ , clearly not symmetric if  $K(x, t)$  alone is symmetric. However, if we multiply Eq. (21.62) by  $\sqrt{\rho(x)}$  and substitute

$$\sqrt{\rho(x)} \varphi(x) = \psi(x),$$

we obtain

$$\psi(x) = \sqrt{\rho(x)} f(x) + \lambda \int_a^b \left[ K(x, t) \sqrt{\rho(x) \rho(t)} \right] \psi(t) dt, \quad (21.63)$$

with a symmetric total kernel  $K(x, t) \sqrt{\rho(x) \rho(t)}$ .

### Orthogonal Eigenfunctions

We now focus on the homogeneous Fredholm equation of the second kind:

$$\varphi(x) = \lambda \int_a^b K(x, t) \varphi(t) dt. \quad (21.64)$$

We assume that the kernel  $K(x, t)$  is symmetric and real. Perhaps one of the first questions we might ask about the equation is: “Does it make sense?” or more precisely, “Does an eigenvalue  $\lambda$  satisfying this equation exist?” This question can be answered in the affirmative. Courant and Hilbert (in their work cited in the Additional Readings, chapter III, section 4) show that if  $K(x, t)$  is continuous, there is at least one such eigenvalue and possibly an infinite number of them.

It is useful to recognize that Eq. (21.64) represents a linear-operator eigenvalue problem: The integral on its right-hand side converts  $\varphi$  into (in general) some other function, which we can indicate symbolically by the equation

$$\psi(x) = \int_a^b K(x, t)\varphi(t) dt \equiv \mathcal{K}\varphi(x), \quad (21.65)$$

so our eigenvalue problem is

$$\mathcal{K}\varphi(x) = \frac{1}{\lambda}\varphi(x). \quad (21.66)$$

We do not have to worry about the possibility that  $\lambda = 0$ , since we can read directly from Eq. (21.64) that in that case the solution to our integral equation will be uniquely  $\varphi(x) = 0$ .

The integral operator  $\mathcal{K}$  is **linear**, since it is obviously true that

$$\mathcal{K}(a\varphi_1(x) + b\varphi_2(x)) = a\mathcal{K}\varphi_1(x) + b\mathcal{K}\varphi_2(x).$$

In addition, if we define the scalar product as an integral on the range  $(a, b)$ :

$$\langle \psi | \varphi \rangle \equiv \int_a^b \psi^*(x)\varphi(x)dx, \quad (21.67)$$

we then see that our requirement that the kernel  $K(x, t)$  be real and symmetric will make  $\mathcal{K}$  a self-adjoint operator:

$$\begin{aligned} \langle \psi | \mathcal{K}\varphi \rangle &= \int_a^b \psi^*(x) \left[ \int_a^b K(x, t)\varphi(t) dt \right] dx = \int_a^b dt \left[ \int_a^b dx K(t, x)\psi(x) \right]^* \varphi(t) \\ &= \langle \mathcal{K}\psi | \varphi \rangle. \end{aligned} \quad (21.68)$$

The linearity and self-adjointness indicate that we can expect to confirm that  $\mathcal{K}$  has the key properties of self-adjoint operators, namely that its eigenvalues are real and (except in the case of degeneracy) its eigenvectors are orthogonal.

While the above constitutes a complete demonstration of the orthogonality of our solutions to the homogeneous Fredholm equation, let's confirm these properties more explicitly.



We can start from the two equations,

$$\frac{1}{\lambda_i} \varphi_i(x) = \int_a^b K(x, t) \varphi_i(t) dt, \quad (21.69)$$

$$\frac{1}{\lambda_j} \varphi_j(x) = \int_a^b K(x, t) \varphi_j(t) dt. \quad (21.70)$$

If we multiply Eq. (21.69) by  $\varphi_j^*(x)$  and Eq. (21.70) by  $\varphi_i^*(x)$  and then integrate with respect to  $x$ , the two equations become<sup>4</sup>

$$\frac{1}{\lambda_i} \int_a^b \varphi_j^*(x) \varphi_i(x) dx = \int_a^b \int_a^b K(x, t) \varphi_j^*(t) \varphi_i(x) dt dx, \quad (21.71)$$

$$\frac{1}{\lambda_j} \int_a^b \varphi_i^*(x) \varphi_j(x) dx = \int_a^b \int_a^b K(x, t) \varphi_i^*(t) \varphi_j(x) dt dx. \quad (21.72)$$

Since we have demanded that  $K(x, t)$  be real and symmetric, we may take the complex conjugate of Eq. (21.72) and then interchange the roles of  $x$  and  $t$  in the integral, reaching

$$\frac{1}{\lambda_j^*} \int_a^b \varphi_i(x) \varphi_j^*(x) dx = \int_a^b \int_a^b K(x, t) \varphi_i(x) \varphi_j^*(t) dt dx. \quad (21.73)$$

Subtracting Eq. (21.73) from Eq. (21.71), we obtain

$$\left( \frac{1}{\lambda_i} - \frac{1}{\lambda_j^*} \right) \int_a^b \varphi_j^*(x) \varphi_i(x) dx = 0. \quad (21.74)$$

Just as in our earlier derivation from Sturm-Liouville theory, we conclude that if  $i = j$  the integral in Eq. (21.74) is necessarily nonzero; so  $1/\lambda_i = 1/\lambda_i^*$ , meaning that  $\lambda_i$  must be real. But if  $\lambda_i \neq \lambda_j$ ,

$$\int_a^b \varphi_i^*(x) \varphi_j(x) dx = 0, \quad \lambda_i \neq \lambda_j, \quad (21.75)$$

proving orthogonality. The derivation can also be completed if  $K(x, t)$  is Hermitian, meaning that  $K(t, x) = K^*(x, t)$ . See Exercise 21.4.1. Since we are mostly concerned with real

<sup>4</sup>We assume that the necessary integrals exist. For an example of a simple pathological case, see Exercise 21.4.4.

$K$ , it is appropriate to assume also that  $\varphi$  is real, and for the remainder of this chapter we will often omit the complex conjugate asterisks that occur, for example, in Eq. (21.75).

If the eigenvalue  $\lambda_i$  is **degenerate**,<sup>5</sup> the eigenfunctions for that particular eigenvalue may be orthogonalized by the Gram-Schmidt method (Section 5.2). Our orthogonal eigenfunctions may, of course, be normalized, and we assume that this has been done. The result is

$$\int_a^b \varphi_i^*(x) \varphi_j(x) dx = \delta_{ij}. \quad (21.76)$$

It can be shown that the eigenfunctions of our integral equations form a complete set,<sup>6</sup> in the sense that if a function  $g(x)$  can be generated by the integral

$$g(x) = \int K(x, t) h(t) dt,$$

with  $h(t)$  a piecewise continuous function, then  $g(x)$  can be represented by a series of eigenfunctions,

$$g(x) = \sum_{n=1}^{\infty} a_n \varphi_n(x). \quad (21.77)$$

The series in Eq. (21.77) can be shown to converge uniformly and absolutely.

Let us extend this to the kernel  $K(x, t)$  by asserting that

$$K(x, t) = \sum_{n=1}^{\infty} a_n \varphi_n(t), \quad (21.78)$$

and  $a_n = a_n(x)$ . Substituting into the original integral equation, Eq. (21.64), and using the orthogonality integral, we obtain

$$\varphi_i(x) = \lambda_i a_i(x). \quad (21.79)$$

Therefore, for our homogeneous Fredholm equation of the second kind, the kernel may be expressed in terms of the eigenfunctions and eigenvalues as

$$K(x, t) = \sum_{n=1}^{\infty} \frac{\varphi_n(x) \varphi_n(t)}{\lambda_n}. \quad (21.80)$$

Equation (21.80) is not actually a new result. In the Green's function chapter, Section 10.1, we identified  $K(x, t)$ , there called  $G(x, t)$ , as the Green's function appearing in Eq. (10.30), with the expansion given in Eq. (10.14). However, it is possible that the

<sup>5</sup>As for differential operators, if more than one distinct eigenfunction of Eq. (21.64) corresponds to the same eigenvalue, that eigenvalue is said to be degenerate.

<sup>6</sup>For a proof of this statement, see Courant and Hilbert (1953), chapter III, section 5, in the Additional Readings.

expansion given by Eq. (21.80) may not exist. As an illustration of the sort of pathological behavior that may occur, you are invited to apply this analysis to

$$\varphi(x) = \lambda \int_0^{\infty} e^{-xt} \varphi(t) dt.$$

Compare Exercise 21.4.4.

It should be emphasized that this Hilbert-Schmidt theory is concerned with the establishment of properties of the eigenvalues (real) and eigenfunctions (orthogonality, completeness), properties that may be of great interest and value. The Hilbert-Schmidt theory does **not solve** the homogeneous integral equation for us any more than the Sturm-Liouville theory for differential equations solved the ODEs. The solutions of the integral equation come by the application of techniques such as were introduced in Sections 21.2 and 21.3, or perhaps even by numerical methods.

## Inhomogeneous Integral Equation

We now continue with the Hilbert-Schmidt theory by seeking solutions of the inhomogeneous equation

$$\varphi(x) = f(x) + \lambda \int_a^b K(x, t) \varphi(t) dt. \quad (21.81)$$

We assume that the solutions of the corresponding homogeneous integral equation are already known:

$$\varphi_n(x) = \lambda_n \int_a^b K(x, t) \varphi_n(t) dt, \quad (21.82)$$

the solution  $\varphi_n(x)$  corresponding to the eigenvalue  $\lambda_n$ . Note that at this point we are assuming nothing about  $\lambda$ ; it is a constant that has no specific relationship to the eigenvalues  $\lambda_n$  of the homogeneous integral equation.

We expand both  $\varphi(x)$  and  $f(x)$  in terms of this set of eigenfunctions:

$$\varphi(x) = \sum_{n=1}^{\infty} a_n \varphi_n(x) \quad (a_n \text{ unknown}), \quad (21.83)$$

$$f(x) = \sum_{n=1}^{\infty} b_n \varphi_n(x) \quad (b_n \text{ known}). \quad (21.84)$$

Substituting into Eq. (21.81), we obtain

$$\sum_{n=1}^{\infty} a_n \varphi_n(x) = \sum_{n=1}^{\infty} b_n \varphi_n(x) + \lambda \int_a^b K(x, t) \sum_{n=1}^{\infty} a_n \varphi_n(t) dt. \quad (21.85)$$

By interchanging the order of integration and summation, we may evaluate the integral by Eq. (21.82), and we get

$$\sum_{n=1}^{\infty} a_n \varphi_n(x) = \sum_{n=1}^{\infty} b_n \varphi_n(x) + \lambda \sum_{n=1}^{\infty} \frac{a_n \varphi_n(x)}{\lambda_n}. \quad (21.86)$$

If we multiply by  $\varphi_i(x)$  and integrate from  $x = a$  to  $x = b$ , the orthogonality of our eigenfunctions leads to

$$a_i = b_i + \lambda \frac{a_i}{\lambda_i}. \quad (21.87)$$

This can be rewritten as

$$a_i = b_i + \frac{\lambda}{\lambda_i - \lambda} b_i. \quad (21.88)$$

We now multiply Eq. (21.88) by  $\varphi_i(x)$  and sum over  $i$ , giving

$$\begin{aligned} \varphi(x) &= f(x) + \lambda \sum_{i=1}^{\infty} \frac{\varphi_i(x)}{\lambda_i - \lambda} b_i \\ &= f(x) + \lambda \sum_{i=1}^{\infty} \frac{\varphi_i(x)}{\lambda_i - \lambda} \int_a^b f(t) \varphi_i(t) dt. \end{aligned} \quad (21.89)$$

Here it is assumed that the eigenfunctions  $\varphi_i(x)$  are normalized to unity. **Note that if  $f(x) = 0$ , there is no solution unless  $\lambda$  is equal to one of the  $\lambda_i$ , thereby confirming that the homogeneous integral equation has only the solutions  $\varphi_i(x)$ .**

In the event that  $\lambda$  for the inhomogeneous equation, Eq. (21.81), is equal to one of the eigenvalues  $\lambda_p$  of the homogeneous equation, our solution, Eq. (21.89), blows up. It can be shown that the inhomogeneous equation then has no solution unless the coefficient  $b_p$  vanishes, meaning that there is no solution unless the inhomogeneous term  $f(x)$  is orthogonal to the eigenfunction  $\varphi_p$ . If the eigenvalue  $\lambda_p$  is degenerate, there will be no solution unless  $f(x)$  is orthogonal to all the degenerate eigenfunctions.

For the case that  $b_p = 0$ , we can return to Eq. (21.87), which then reduces for  $a_p$  to

$$a_p = b_p + a_p = a_p, \quad (21.90)$$

which gives no information about  $a_p$ . Note that if  $b_p \neq 0$  this equation cannot be satisfied, a signal that a solution cannot be obtained.

Under the assumption that  $b_p = 0$  we now can rewrite Eq. (21.86), identifying its first two summations, respectively, as  $\varphi(x)$  and  $f(x)$ , separating the final summation into the single term  $a_p \varphi_p(x)$  plus a sum over all  $n$  other than  $p$ , thereby reaching

$$\varphi(x) = f(x) + a_p \varphi_p + \lambda_p \sum_{i=1}^{\infty} \frac{\varphi_i(x)}{\lambda_i - \lambda_p} \int_a^b f(t) \varphi_i(t) dt. \quad (21.91)$$

In this solution the  $a_p$  remains as an undetermined constant,<sup>7</sup> and the prime indicates that  $i = p$  is to be omitted from the sum.

It is of interest to relate Eq. (21.89) to what might be expected if we tried to develop a similar equation by Green's-function methods. To do this, we start by rewriting Eq. (21.81) as an operator equation of the form

$$\mathcal{K}\varphi(x) - \frac{1}{\lambda}\varphi(x) = -\frac{f(x)}{\lambda}, \quad (21.92)$$

where  $\mathcal{K}$  is the operator that we introduced in Eq. (21.65). Next, we note that, from Eq. (21.82), the  $\varphi_n$  are eigenfunctions of  $\mathcal{K}$  with eigenvalues  $1/\lambda_n$ :

$$\mathcal{K}\varphi_n(x) = \frac{\varphi_n(x)}{\lambda_n}. \quad (21.93)$$

Then, applying Eq. (10.39), the Green's function of the entire left-hand side of Eq. (21.92) will be (assuming  $\varphi$  is real):

$$\begin{aligned} G(x, t) &= \sum_n \frac{\varphi_n(x)\varphi_n(t)}{\lambda_n^{-1} - \lambda^{-1}} = \lambda \sum_n \frac{\lambda_n}{\lambda - \lambda_n} \varphi_n(x)\varphi_n(t) \\ &= -\lambda \sum_n \varphi_n(x)\varphi_n(t) + \lambda^2 \sum_n \frac{\varphi_n(x)\varphi_n(t)}{\lambda - \lambda_n} \\ &= -\lambda\delta(t - x) - \lambda^2 \sum_n \frac{\varphi_n(x)\varphi_n(t)}{\lambda_n - \lambda}. \end{aligned} \quad (21.94)$$

To reach the last line of Eq. (21.94) we used the eigenfunction expansion of the delta function, Eq. (5.27). Applying this Green's function to the right-hand side of Eq. (21.92), we get

$$\begin{aligned} \varphi(x) &= -\frac{1}{\lambda} \int_a^b G(x, t) f(t) dt \\ &= -\frac{1}{\lambda} \int_a^b \left[ -\lambda\delta(t - x) - \lambda^2 \sum_n \frac{\varphi_n(x)\varphi_n(t)}{\lambda_n - \lambda} \right] f(t) dt \\ &= f(x) + \lambda \sum_n \frac{\varphi_n(x)}{\lambda_n - \lambda} \int_a^b \varphi_n(t) f(t) dt, \end{aligned} \quad (21.95)$$

which agrees with Eq. (21.89).

<sup>7</sup>This is like the inhomogeneous linear ODE. We may add to its solution any constant times a solution of the corresponding homogeneous ODE.

**Example 21.4.1** INHOMOGENEOUS FREDHOLM EQUATION

Let's seek solutions to the inhomogeneous Fredholm equation

$$\varphi(x) = x^3 + \lambda \int_{-1}^1 (t+x)\varphi(t) dt, \quad (21.96)$$

for the two  $\lambda$  values  $\lambda = 1$  and  $\lambda = \sqrt{3}/2$ . The corresponding homogeneous equation, treated in [Example 21.2.3](#), has solutions only for the two eigenvalues  $\pm\sqrt{3}/2$ . In normalized form, they are:

$$\lambda_1 = \frac{\sqrt{3}}{2}, \quad \varphi_1 = \frac{\sqrt{3}}{2} \left( x + \frac{1}{\sqrt{3}} \right); \quad \lambda_2 = -\frac{\sqrt{3}}{2}, \quad \varphi_2 = \frac{\sqrt{3}}{2} \left( x - \frac{1}{\sqrt{3}} \right).$$

Taking first  $\lambda = 1$ , which is not an eigenvalue of the homogeneous equation, we have

$$\begin{aligned} \varphi(x) &= x^3 + \sum_{i=1}^2 \frac{\varphi_i(x)}{\lambda_i - 1} \int_{-1}^1 t^3 \varphi_i(t) dt \\ &= x^3 + \frac{\frac{\sqrt{3}}{2} \left( x + \frac{1}{\sqrt{3}} \right)}{\frac{\sqrt{3}}{2} - 1} \frac{\sqrt{3}}{2} \int_{-1}^1 t^3 \left( t + \frac{1}{\sqrt{3}} \right) dt \\ &\quad + \frac{\frac{\sqrt{3}}{2} \left( x - \frac{1}{\sqrt{3}} \right)}{-\frac{\sqrt{3}}{2} - 1} \frac{\sqrt{3}}{2} \int_{-1}^1 t^3 \left( t - \frac{1}{\sqrt{3}} \right) dt \\ &= x^3 - \frac{6}{5}(2x+1). \end{aligned} \quad (21.97)$$

Continuing now to  $\lambda = \sqrt{3}/2$ , we note that it is the eigenvalue  $\lambda_1$  of the homogeneous integral equation. That means the integral equation will have no solution unless  $\langle \varphi_1 | f \rangle = 0$ . For the present problem,

$$\langle \varphi_1 | f \rangle = \frac{\sqrt{3}}{2} \int_{-1}^1 \left( x + \frac{1}{\sqrt{3}} \right) x^3 dx \neq 0,$$

so our integral equation will have no solution for  $\lambda = \sqrt{3}/2$ . If in spite of this observation we attempted to generate a solution using [Eq. \(21.91\)](#), the function  $\varphi(x)$  we obtained would not satisfy the integral equation, irrespective of the value we might choose to assign to  $a_p$ . The immediate reason we cannot obtain a solution is that the integral

$$\int_{-1}^1 (t+x)f(t)dt = \int_{-1}^1 (t+x)t^3 dt = \frac{2}{5}$$

evaluates to a quantity that cannot be represented as a linear combination of the eigenfunctions  $\varphi_i$  other than  $\varphi_p$  (in the present case, this means that  $2/5$  is not proportional to  $\varphi_2$ ). There is therefore no way to add an additional component to  $f(x)$  to obtain a cancellation of the  $2/5$ . ■

## Exercises

**21.4.1** In the Fredholm equation

$$\varphi(x) = \lambda \int_a^b K(x, t) \varphi(t) dt,$$

assume that the kernel  $K(x, t)$  is self-adjoint or Hermitian:

$$K(x, t) = K^*(t, x).$$

Extend the analysis of the present section to show that

(a) the eigenfunctions are orthogonal, in the sense that

$$\int_a^b \varphi_m^*(x) \varphi_n(x) dx = 0, \quad m \neq n (\lambda_m \neq \lambda_n).$$

(b) the eigenvalues are real.

**21.4.2** (a) Show that the eigenfunctions of [Exercise 21.2.12](#) are orthogonal.

(b) Show that the eigenfunctions of [Exercise 21.2.14](#) are orthogonal.

**21.4.3** Use the Hilbert-Schmidt method to solve the inhomogeneous integral equation

$$\varphi(x) = x + \frac{1}{2} \int_{-1}^1 (t + x) \varphi(t) dt.$$

The corresponding homogeneous integral equation was treated in [Example 21.2.3](#).

*Note.* The application of the Hilbert-Schmidt technique here is somewhat like using a shotgun to kill a mosquito, especially when the equation can be solved quickly by expanding in Legendre polynomials.

**21.4.4** The Fredholm integral equation

$$\varphi(x) = \lambda \int_0^\infty e^{-xt} \varphi(t) dt$$

has an infinite number of solutions, of which one is

$$\varphi(x) = x^{-1/2}, \quad \lambda = \pi^{-1/2}.$$

Verify that this is a solution and that it is **not** normalizable.

*Note.* A basic reason for this anomalous behavior is that the range of integration is infinite, making this a “singular” integral equation. Note also that a series expansion of the kernel  $e^{-xt}$  would permit a solution by the separable-kernel method (Section 21.2), except that the series is infinite. This observation is consistent with the fact that this integral equation has an infinite number of eigenvalues and eigenfunctions.

**21.4.5** Given

$$y(x) = x + \lambda \int_0^1 xt \, y(t) \, dt:$$

- (a) Determine  $y(x)$  as a Neumann series.
- (b) Find the range of  $\lambda$  for which your Neumann series solution is convergent. Compare with the value obtained from

$$|\lambda| \cdot |K|_{\max} < 1.$$

- (c) Find the eigenvalue and the eigenfunction of the corresponding homogeneous integral equation.
- (d) By the separable-kernel method show that the solution is

$$y(x) = \frac{3x}{3 - \lambda}.$$

- (e) Find  $y(x)$  by the Hilbert-Schmidt method.

**21.4.6** In Exercise 21.2.11 it was found that the integral equation

$$\varphi(x) = \lambda \int_0^{2\pi} \cos(x - t) \varphi(t) \, dt$$

had (unnormalized) eigenfunctions  $\cos x$  and  $\sin x$ , both with eigenvalue  $\lambda_i = 1/\pi$ . Show that the kernel of this integral equation has an expansion of the form

$$K(x, t) = \sum_{n=1}^2 \frac{\varphi_n(x) \varphi_n(t)}{\lambda_n}.$$

**21.4.7** The integral equation  $\varphi(x) = \lambda \int_0^1 (1 + xt) \varphi(t) \, dt$

has eigenvalues  $\lambda_1 = 0.7889$  and  $\lambda_2 = 15.211$ . The corresponding eigenfunctions are  $\varphi_1 = 1 + 0.5352x$  and  $\varphi_2 = 1 - 1.8685x$ .

- (a) Show that these eigenfunctions are orthogonal over the interval  $[0, 1]$ .
- (b) Normalize the eigenfunctions to unity.



(c) Show that

$$K(x, t) = \frac{\varphi_1(x)\varphi_1(t)}{\lambda_1} + \frac{\varphi_2(x)\varphi_2(t)}{\lambda_2}.$$

ANS. (b)  $\varphi_1(x) = 0.7831 + 0.4191x,$   
 $\varphi_2(x) = 1.8403 - 3.4386x.$

**21.4.8** An alternate form of the solution to the inhomogeneous integral equation, Eq. (21.81), is

$$\varphi(x) = \sum_{i=1}^{\infty} \frac{b_i \lambda_i}{\lambda_i - \lambda} \varphi_i(x).$$

- (a) Derive this form without using Eq. (21.89).  
 (b) Show that this form and Eq. (21.89) are equivalent.

### Additional Readings

- Bocher, M., *An Introduction to the Study of Integral Equations*, Cambridge Tracts in Mathematics and Mathematical Physics, No. 10. New York: Hafner (1960). This is a helpful introduction to integral equations.
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- Cochran, J. A., *The Analysis of Linear Integral Equations*. New York: McGraw-Hill (1972). This is a comprehensive treatment of linear integral equations intended for applied mathematicians and mathematical physicists. It assumes a moderate to high level of mathematical competence on the part of the reader.
- Courant, R., and D. Hilbert, *Methods of Mathematical Physics*, Vol. 1 (English edition). New York: Interscience (1953). This is one of the classic works of mathematical physics. Originally published in German in 1924, the revised English edition is an excellent reference for a rigorous treatment of integral equations, Green's functions, and a wide variety of other topics on mathematical physics.
- Golberg, M. A., ed., *Solution Methods of Integral Equations*. New York: Plenum Press (1979). This is a set of papers from a conference on integral equations. The initial chapter is excellent for up-to-date orientation and a wealth of references.
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